

ToyADMOS2026: A Dual-Channel Dataset for Noise-Aware First-Shot Unsupervised Anomalous Sound Detection

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Abstract—Unsupervised anomalous sound detection for machine condition monitoring degrades when the background noise is non-stationary, leading to a mismatch between training and test conditions. DCASE 2026 Task 2, therefore, introduced noise-aware ASD with synchronized near/far microphone signals, while retaining the first-shot and domain-generalization requirements of DCASE 2025 Task 2. Because existing ASD datasets did not provide this observation, we recorded ToyCar, ToyDrone, and ToothBrush, and generated ToyCarEmu using measured impulse responses. The microphone arrangement is fixed within each machine type, and the two channels provide different target-to-noise dominance and arrival-time cues. We also adapted the DCASE 2025 baseline to stereo WAV input; it uses only the near channel as a reference for two-channel systems. We describe the four machine-type recordings, baseline, and evaluation results, providing a reproducible benchmark for noise-aware first-shot ASD.

Index Terms—anomalous sound detection, machine condition monitoring, noise-aware learning, dual-channel recording, first-shot learning

1. INTRODUCTION

Unsupervised anomalous sound detection (ASD) detects unknown machine faults from models trained only on normal operating sounds. DCASE Task 2 established this setting in 2020 with ToyADMOS and MIMII recordings of damaged miniature and industrial machines [1]–[3]. In practice, however, a target machine is observed together with surrounding equipment and environmental noise. When background noise is non-stationary, it can cause a mismatch between training and test conditions, resemble an anomaly, and degrade machine-condition monitoring.

The task evolved to address increasingly realistic distribution shifts. The 2021 edition introduced domain shift, and the 2022 edition introduced domain generalization [4], [5], supported by ToyADMOS2, MIMII DUE, and MIMII DG [6]–[8]. First-shot ASD followed in 2023 and continued through 2025 [9]–[13], using ToyADMOS2+, ToyADMOS2#, and ToyADMOS2025 [14]–[16]. Development and evaluation use different machine types with source-to-target shifts, and evaluation-specific tuning is prohibited.

DCASE 2026 Task 2 retains these first-shot requirements and introduces noise-aware ASD [17], [18]. Each clip provides synchronized observations from a near microphone, dominated by the target, and a far microphone, more informative about the background. The far channel is not a noise-only reference: both microphones capture mixtures through different propagation paths. Their relative level, delay, phase, and learned cross-channel relationships can help separate machine variation from environmental change on practical stereo-input IoT devices. Existing ASD datasets, however, lacked synchronized near/far recordings with this controlled geometry, requiring new real recordings and impulse-response (IR)-based emulations.

This paper makes three contributions: (1) it formalizes the noise-aware near/far setting; (2) it documents the recording, generation, operating conditions, domain shifts, and physical anomalies of ToyCar, ToyCarEmu, ToyDrone, and ToothBrush; and (3) it presents the stereo-

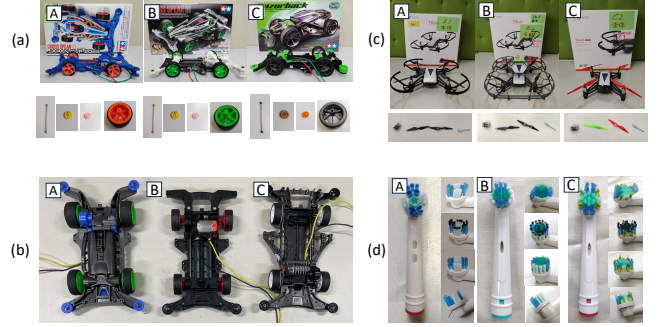


Fig. 1: Target-machine configurations for (a) ToyCar, (b) ToyCarEmu, (c) ToyDrone, and (d) ToothBrush.

Table 1: Dataset components contributed by the authors.

Machine	Phase	Type	Creation and metadata
ToyCar	Dev.	Real	Newly assembled cars; attributes hidden.
ToyCarEmu	Dev.	Emu.	ToyADMOS2 recordings and measured IRs; attributes provided.
ToyDrone	Add./Eval.	Real	Hovering drones; attributes provided for training.
ToothBrush	Add./Eval.	Real	Fixed-point brushing; attributes hidden.

compatible DCASE 2026 baseline¹ and analyzes its results alongside a top-ranked two-channel system.

2. NOISE-AWARE FIRST-SHOT ASD TASK

The task remains unsupervised: a system learns from normal clips and assigns an anomaly score to each test clip. Its development and evaluation machine types differ, and each machine type includes a source-to-target domain shift. Model selection on development machines must therefore generalize without tuning to the evaluation machine types.

The new input is $\mathbf{x} \in \mathbb{R}^{2 \times L}$. In the stereo WAV files provided in the dataset, the left channel is the near signal, and the right channel is the far signal. These file-channel indices must be distinguished from the physical recorder microphone numbers in Table 2. Both channels contain the target and environmental sounds, but with different propagation paths and target-to-noise dominance. This design permits single-channel, noise-reference, and fully multi-channel approaches.

It should be noted that this two-channel setting is easily implemented using an ordinary IoT device with a stereo microphone input. Besides, locating two microphones, one closer to the target machine and the other farther from the target or closer to the noise source, is feasible without much extra cost. Therefore, if a system performs well in this task setting, the solution should generalize to realistic scenarios in noisy environments for machine condition monitoring.

¹https://github.com/nttcs/nttcs-dcase2023_task2_baseline_ae/releases/tag/v5.3.0

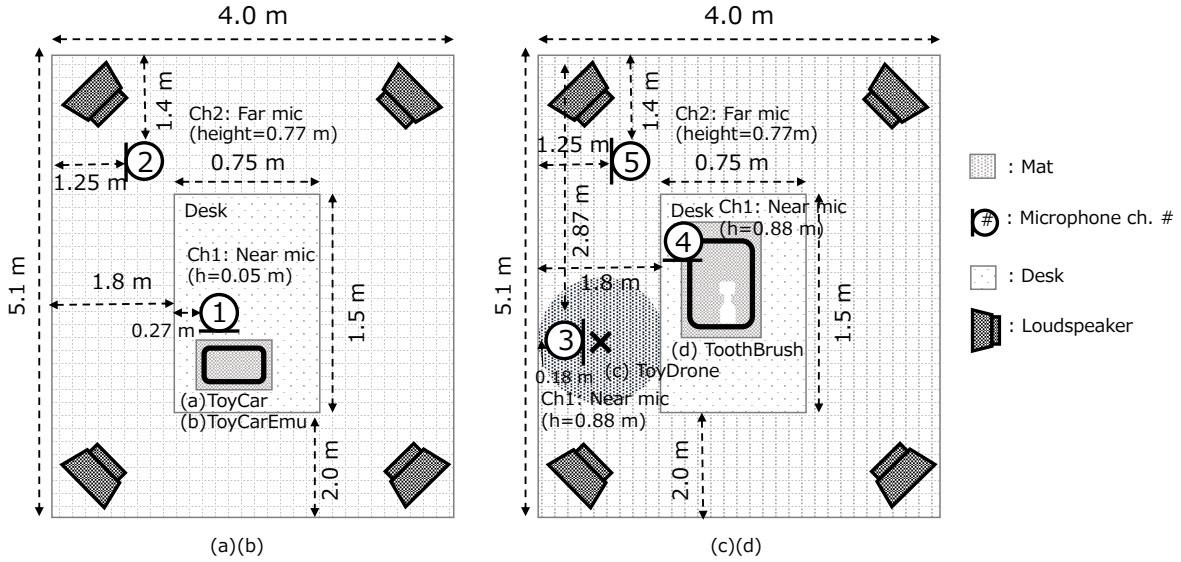


Fig. 2: Room layouts and microphone arrangements for (a) ToyCar, (b) ToyCarEmu, (c) ToyDrone, and (d) ToothBrush. Ch. 1/2 denote the distributed near/far channels; Table 2 lists the physical recording microphones.

Table 2: Conditions of the four ToyADMOS2026 datasets, including non-stationary background noise that can cause a mismatch between training and test conditions. Mic. gives physical recording-microphone numbers, not distributed WAV channel indices.

Machine	Settings and parameter variations					Domain shift		Samples	
	ID	Dur.	Mic. (near, far)	Speed/mode	Background noise	Source→target		Training	Test
ToyCar	A, B, C	11 s	Ch. 1, 2	1: 2.8 V; 2: 3.1 V; 3: 3.4 V; 4: 3.7 V	Nihonbashi Bridge underpass	ID: A, B→C; Speed: 1,2,3→1,4		Src. 990 Trg. 10	Src. 100 Trg. 100
ToyCarEmu	A, B, C	12 s	Ch. 1, 2	1: 2.8 V; 2: 3.1 V; 3: 3.4 V; 4: 3.7 V	Nihonbashi Bridge underpass	ID: A, B→C; Speed: 1, 2, 3→1, 4		Src. 990 Trg. 10	Src. 100 Trg. 100
ToyDrone	A, B, C	16 s	Ch. 3, 5	1: fast; 2: mid; 3: slow	Park drainage water	ID: A, B→C; Speed: 1,2→3		Src. 990 Trg. 10	Src. 100 Trg. 100
ToothBrush	A, B, C	6 s	Ch. 4, 5	mode 1: 2.8–3.1 V; mode 2: 2.7–3.2 V	Bathroom-air-conditioner and fan	ID: A, B→C; mode: 1→2		Src. 990 Trg. 10	Src. 100 Trg. 100

3. DATASET OVERVIEW

The complete development set contains ToyCar, ToyCarEmu, Fan, GearboxEmu, BearingEmu, SliderEmu, and ValveEmu. The additional training and evaluation sets contain ToyDrone, ToothBrush, SewingMachine, Sander, and BlowerDustCollector [19]–[21]. This paper focuses on the four configurations in Fig. 1 and Table 1 contributed by the authors. In the evaluation test set, normal/anomaly, source/target, and attribute labels are hidden for every machine type.

ToyCar, ToyDrone, and ToothBrush are real two-microphone recordings. Datasets whose names contain “Emu” combine existing machine recordings and environmental noise after convolution with newly measured transfer paths. In particular, the ToyADMOS2 signal used for ToyCarEmu is an already-recorded near-equivalent microphone signal, not a dry or anechoic machine signal [6]. ToyCarEmu is thus not a duplicated-noise stereo signal, but a two-channel mixture produced with measured propagation characteristics.

For all four datasets, the microphone geometry is fixed between training and test within a machine type. Original signals and IRs were recorded at 48 kHz and 24 bit. After any IR convolution, RMS-based gain adjustment, and channel-wise mixing, the released clips were formatted as 16 kHz, 24-bit, two-channel WAV files.

The mixing SNRs are calculated based on the near-channel

contribution by

$$SNR = 20 \log_{10} \left\{ \frac{1}{J} \sum rms(S_j[near]) \right\} / rms(N_k[near]), \quad (1)$$

where $S_j[near]$ is a recorded machine-operating sound sample in the near channel and $N_k[near]$ is a recorded environmental noise sample with the same duration as $S_j[near]$, were set to 6 dB for ToyCar, 3 dB for ToyCarEmu, 12 dB for ToyDrone, and −20 dB for ToothBrush in terms of RMS values. The same gain factor for the noise sample in the near channel is applied to the noise sample in the far channel. After being mixed with the noise, all sound samples were downsampled to 16 kHz.

4. RECORDING AND GENERATION

4.1. ToyCar and ToyCarEmu for Dev. set

As shown in Figs. 2(a), (b) and 3(a), (b), a bench was placed on a table near the center of an approximately 4.0 m × 5.1 m room. The near microphone was positioned beside the Mini4WD at a height of 0.05 m, and the far microphone was separated from it at a height of 0.77 m. Four loudspeakers near the room corners reproduced environmental sounds recorded near the Nihonbashi Bridge underpass. Machine operation and loudspeaker noise were captured separately with the same near/far geometry, adjusted using their recorded RMS values, and mixed channel by channel. We intentionally describe this operation as

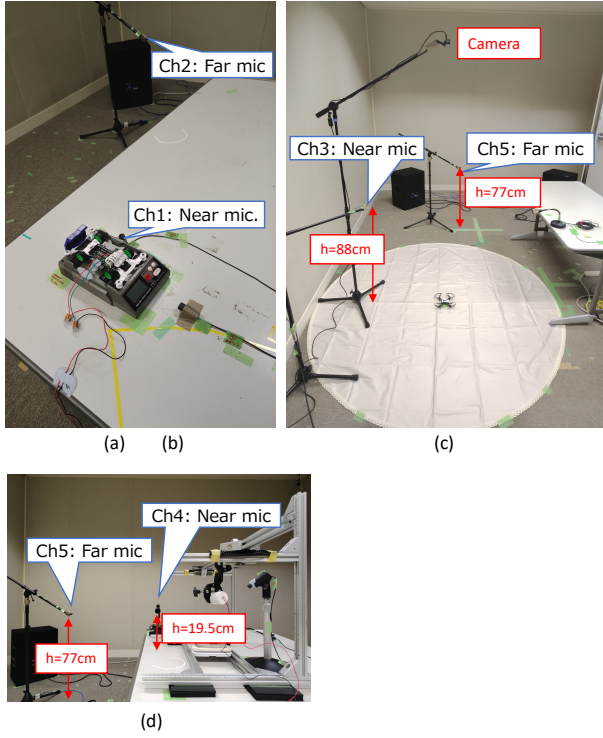


Fig. 3: The near/far microphones for (a) ToyCar, (b) ToyCarEmu, (c) ToyDrone, and (d) ToothBrush.

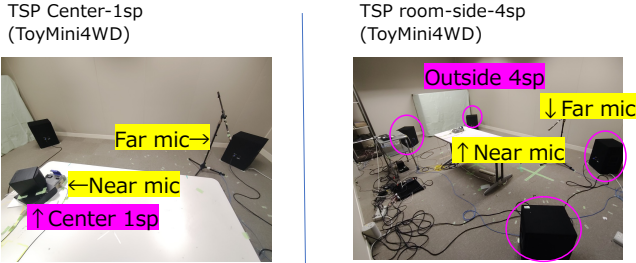


Fig. 4: IR measurement for ToyCarEmu from the target and environmental-noise loudspeaker positions to the near/far microphones.

RMS-based gain adjustment rather than assigning a single S/N ratio to nonstationary signals.

ToyCar uses newly assembled machine instances and direct near/far recordings. Its source-to-target changes cover machine ID and operating voltage/speed, as summarized in Table 2. Anomalies were physically introduced in the shaft, gears, or wheels (Table 3).

ToyCarEmu instead starts from the near-equivalent ToyCar recordings distributed in ToyADMOS2. Fig. 4 shows the newly measured IRs from the target position and each environmental-noise loudspeaker position to both microphones. The corresponding machine and noise signals were convolved with these IRs, adjusted using their RMS values, and mixed independently for the two channels. ToyCar and ToyCarEmu share the microphone arrangement and background noise, enabling a comparison between direct recording and IR-based emulation despite their different base-machine recordings.

4.2. ToyDrone and ToothBrush for Add./Eval. set

ToyDrone was operated by hovering vertically, with the near microphone approximately 0.65 m from the drone center and environmental water noise reproduced from the room loudspeakers [Figs. 1(c),

Table 3: Physical anomaly conditions.

Machine	Part	Condition
ToyCar / ToyCarEmu	Shaft	Bent
	Gears	Deformed; melted
	Wheels	Damaged
ToyDrone	Propeller	Offset weight; weights on two sides
	Arm	Offset weight
	Unbalance	Offset weight
ToothBrush	Brush hair	Damaged; foreign object; partially missing
	Brush head	Foreign object stuck

2(c), and 3(c)). Its domain shift changes the drone’s instance and speed; anomalous conditions introduce imbalance in the propeller, arm, or body. ToothBrush repeatedly brushed a fixed point while air-conditioner and fan noise were reproduced [Figs. 1(d), 2(d), and 3(d)]. Its domain shift changes the toothbrush instance and mode, and anomalies modify the brush hair or head. Table 3 gives the physical conditions.

5. ORGANIZATION AND METADATA

Each machine type has one section (section 00). Development training contains 990 source-domain and 10 target-domain normal clips; its test set contains 50 normal and 50 anomalous clips in each domain. Additional training provides the same normal training counts for unseen machine types. Evaluation provides 200 test clips per machine type with all test labels hidden. File names and attribute CSVs provide the phase-appropriate machine, domain, and attribute information. ToyCarEmu and ToyDrone have explicit training attributes, whereas the attributes of ToyCar and ToothBrush are hidden.

Development, additional-training, and evaluation packages are published separately on Zenodo [19]–[21]. After the challenge, the evaluation ground truth, official results, participating system reports, and the baseline implementation were also made public [18], [23]. Version 4.x.x of the repository supports DCASE2025T2, whereas version 5.x.x supports DCASE2026T2. Together with the recording details above, these resources support independent reproduction and analysis.

6. BASELINE AND RESULTS

6.1. Stereo-Compatible Baseline

The DCASE 2026 baseline inherits the two anomaly-scoring modes used previously: a first-shot-compliant autoencoder with mean-squared reconstruction error and an autoencoder with selective Mahalanobis scoring [23], [24]. Its loader accepts stereo WAV files with left/near and right/far signals. It then selects only the left channel and applies the same feature extraction, training, and scoring as the DCASE 2025 baseline. Thus, version 5.x.x supports the new file format and protocol, but intentionally provides a near-only reference that does not exploit the noise information in the far channel.

The baseline has the following two operating modes:

First-shot-compliant simple Autoencoder mode (FS-AE): This is a simple autoencoder. For training, the model parameter θ of the AE is trained to minimize the mean square error (MSE) between the near channel of a normal input sample $x^-[\text{near}]$ and its reconstruction $\hat{x}^-[\text{near}]$ using

$$\text{Loss} = \text{MSE}(x^-[\text{near}], \hat{x}^-[\text{near}]), \quad (2)$$

$$\text{where } \hat{x}^-[\text{near}] = \text{Dec}_\theta(\text{Enc}_\theta(x^-[\text{near}])). \quad (3)$$

For the testing phase, the anomaly score A_θ is calculated with the reconstruction error of the given query sample x using

$$\text{Anomaly Score } A_\theta = \text{MSE}(x[\text{near}], \hat{x}[\text{near}]), \quad (4)$$

$$\text{where } \hat{x}[\text{near}] = \text{Dec}_\theta(\text{Enc}_\theta(x[\text{near}])). \quad (5)$$

Table 4: Official DCASE 2026 Task 2 evaluation results. Values are fractions; hmean and amean aggregate the listed AUC/pAUC values.

System	Metric	hmean	amean	ToyDrone	ToothBrush	SewingMachine	Sander	BlowerDustCollector	ToyCar*	ToyCarEmu*
FS-AE (near only)	AUC (source)	0.6606	0.6635	0.7126	0.5902	0.6484	0.6714	0.6948	0.7562	0.6962
	AUC (target)	0.5732	0.6099	0.4680	0.6434	0.6142	0.4402	0.8838	0.3787	0.6120
	pAUC	0.5687	0.5714	0.5511	0.5979	0.5811	0.5089	0.6179	0.5403	0.5589
	Total	0.5980	0.6149						0.5160	0.6173
Selective Mahalanobis AE (near only)	AUC (source)	0.6086	0.6291	0.7276	0.4398	0.6034	0.6980	0.6766	0.7728	0.6949
	AUC (target)	0.4985	0.5395	0.5428	0.3940	0.4376	0.4510	0.8720	0.5317	0.6662
	pAUC	0.5467	0.5520	0.5926	0.5279	0.5058	0.4947	0.6389	0.5825	0.5347
	Total	0.5476	0.5735						0.6133	0.6237
Fujimura_MERL_task2_3 (near/far)	AUC (source)	0.7381	0.7401	0.7252	0.6930	0.7492	0.7220	0.8112	–	–
	AUC (target)	0.7279	0.7431	0.7108	0.6744	0.6828	0.6704	0.9770	–	–
NA-SSL ensemble [22]	pAUC	0.6484	0.6543	0.6353	0.6174	0.6879	0.5737	0.7574	–	–
	Total	0.7024	0.7125						0.5414	0.5780

* ToyCar and ToyCarEmu are results on the development set and are excluded from hmean and amean.

When the anomaly score exceeds the preset threshold, the sample is detected as an anomaly.

Selective Mahalanobis Autoencoder mode: The anomaly score A_θ is calculated using the covariance matrixes Σ_s^{-1} and Σ_t^{-1} of distance between the near channel of normal samples $x^-[\text{near}]$ and its reconstruction $\hat{x}^-[\text{near}]$ for the source and target domains with:

$$\text{Anomaly Score } A_\theta = \min\{D_s(x[\text{near}], \hat{x}[\text{near}]), D_t(x[\text{near}], \hat{x}[\text{near}])\}, \quad (6)$$

$$\text{where } D_s(\cdot) = \text{Mahalanobis}(x[\text{near}], \hat{x}[\text{near}], \Sigma_s^{-1}), \quad (7)$$

$$D_t(\cdot) = \text{Mahalanobis}(x[\text{near}], \hat{x}[\text{near}], \Sigma_t^{-1}). \quad (8)$$

6.2. Evaluation Metrics and Ranking

Participants submit an anomaly score and a binary decision for each test clip. Anomaly scores determine source/target area under the ROC curve (AUC), partial AUC (pAUC) for false-positive rates up to 0.1, and the official harmonic-mean score; decisions yield precision, recall, and F1 but do not affect ranking [18]. Table 4 reports the published evaluation results as fractions. The system hmean is the official ranking statistic; amean is included for reference.

The total score Ω for evaluating the systems is calculated based on Area Under the Receiver Operating Characteristic (ROC) curve (AUC) and partial AUC (pAUC) with a harmonic mean (*hmean*) of AUC and pAUC using the following formula:

$$\Omega = \text{hmean}\{AUC_{m,n,d}, pAUC_{m,n} \mid m \in \mathcal{M}, n \in \mathcal{S}(m), d \in \{\text{source}, \text{target}\}\}, \quad (9)$$

$$AUC_{m,n,d} = \frac{1}{N_d^- N_n^+} \sum_{i=1}^{N_d^-} \sum_{j=1}^{N_n^+} \mathcal{H}(A_\theta(x_j^+) - A_\theta(x_i^-)), \quad (10)$$

$$pAUC_{m,n} = \frac{1}{\lfloor pN_n^- \rfloor N_n^+} \sum_{i=1}^{\lfloor pN_n^- \rfloor} \sum_{j=1}^{N_n^+} \mathcal{H}(A_\theta(x_j^+) - A_\theta(x_i^-)). \quad (11)$$

The AUC is calculated based on the anomaly scores $A_\theta(\cdot)$ of normal samples x_i^- and anomaly samples x_j^+ on each domain $d \in \{\text{source}, \text{target}\}$ of machine type $m \in \mathcal{M}$ and averaged over sections $n \in \mathcal{S}(m)$, where m is the machine type indicator and n is the section indicator in the machine type.

The pAUC is calculated on each machine type $m \in \mathcal{M}$ and averaged over sections across domains. N_d^- is the number of normal samples in each domain. N_n^- and N_n^+ are the numbers of normal and anomaly samples in the section n of the machine type m , respectively. $\mathcal{H}(\cdot)$ is a function that returns 1 when $x > 0$ and 0 otherwise. For the pAUC calculation, the false-positive rate (FPP) p is set to 0.1. $\lfloor \cdot \rfloor$ is a floor function.

7. DISCUSSION

The near-only baselines ensure continuity with earlier first-shot ASD. Noise-aware systems can instead condition or suppress the near signal using the far channel, or learn inter-channel cues in level, timing, phase, and representation. Because the far microphone captures reverberation and leaked machine sound in addition to environmental noise, it serves as a practical reference rather than an oracle noise signal.

The systems-ranking winner, Fujimura_MERL_task2_3, uses both channels in noise-aware audio self-supervised learning (NA-SSL) and scores anomalies in a denoised representation space [22]. Its higher aggregate result in Table 4 demonstrates the promise of the noise-aware setting. The gain cannot be attributed solely to the second channel, however, because pre-trained models, discriminative fine-tuning, scoring, and ensemble design also differ from the baselines. Controlled near-only versus near/far ablations remain essential.

Finally, stereo capture is available on practical IoT hardware. A method that learns the normal near/far relationship and remains robust to changes in background noise can therefore transfer naturally from this benchmark to industrial monitoring without assuming access to a clean target or a noise-only sensor.

8. CONCLUSION

DCASE 2026 Task 2 introduced a noise-aware first-shot ASD using synchronized near- and far-field inputs. To support it, we provided real recordings of ToyCar, ToyDrone, and ToothBrush, as well as ToyCarEmu generated from ToyADMOS2 recordings and newly measured IRs. ToyCar and ToyCarEmu share the microphone arrangement and background noise, allowing direct and IR-based data construction to be compared. We also released a stereo-compatible baseline that processes only the near channel, providing a reference for methods that exploit both channels. The recording, generation, anomaly, and metadata details documented here, together with the released data, ground truth, code, results, and technical reports, make the benchmark reproducible. The strong results of top-performing DCASE 2026 Task 2 systems, including a near/far NA-SSL system, further motivate the use of noise-aware ASD for practical stereo-input industrial monitoring.

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